

Predicting Salary Bands in the AI Job Market

A supervised classification project that predicts whether an AI/data professional falls into a **Low, Medium or High**-salary band, and tests whether daily use of AI tools pays more.

Business Problem

Compensation in the AI job market varies widely across roles like LLM Engineer, Machine Learning Engineer and Data Analyst, and candidates have little reliable benchmarking. A recurring claim in 2026 hiring discourse is that fluency with AI tooling commands a premium, or, conversely, that it prices workers out.

This project asks two questions:

1. **Can salary band be predicted** from role, seniority, industry and working-pattern attributes?
2. **Does daily AI tool use move the needle** on pay, once seniority is accounted for?

Salary is reframed from a continuous target into three bands because that is how recruiters and candidates actually benchmark, against ranges, not exact figures.

Dataset

Source	Google Dataset Search, <i>AI in Software Development Statistics 2026</i>
File	ai_ds_job_salaries_2026.csv
Size	1,000 rows × 27 columns
Target	salary_usd, binned at the 33rd/67th percentiles
Features used	10 (5 categorical, 5 numeric)
Missing values	0
Duplicates	0

Salary band boundaries: Low \$12,000–\$71,730 · Medium \$71,730–\$117,271 · High \$117,271–\$329,944
Classes are balanced by construction (334 / 332 / 334), giving a **baseline accuracy of 33.5%**.

Approach

Preprocessing

- One-hot encoding categorical features, with handle unknown='ignore' for unseen job titles
- StandardScaler for the distance-based SVM; passthrough for tree models
- All transforms wrapped in a ColumnTransformer + Pipeline so nothing is fitted on test data
- Stratified 80/20 split (800 train / 200 test)

Exploratory analysis

- Distribution and outlier checks on the continuous target
- Pearson correlation for numeric features; **Cramér's V** for categorical association with the salary band
- Pivot table of mean salary across experience level × education level
- Collinearity check between the two AI-usage features

Models

Model	Configuration
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Decision Tree	max_depth=4, min_samples_leaf=5
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Random Forest	200 trees, max_features='sqrt'
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SVM	RBF kernel, C=1.0, gamma='scale'
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Evaluation = accuracy, per-class precision/recall/F1, confusion matrices, one-vs-rest ROC-AUC, 5-fold stratified cross-validation, and an explicit train-vs-test overfitting check.

Results

Model	Test accuracy 5-fold CV	Train	Train-test gap
Decision Tree	0.495	0.536 (± 0.023)	0.570 0.075

Model	Test accuracy 5-fold CV		Train	Train–test gap
Random Forest	0.525	0.520 ($\pm$ 0.038)	1.000	0.475
SVM	0.510	0.528 ($\pm$ 0.021)	0.689	0.179

All three models beat the 33.5% baseline by 17–19 points. Critically, **the single split and Cross-validation disagree on the ranking**, nothing about the models changed, only which rows landed in the test set. All three sit within one standard deviation of each other, so no model is genuinely superior.

The **pruned Decision Tree is the recommended model**: it leads on cross-validation, has by far the smallest train–test gap, and is directly explainable to a non-technical audience. Random Forest scored a perfect 1.000 on training data, textbook memorization.

Selected ROC-AUC (SVM, one-vs-rest): Low 0.720 · Medium 0.575 · High 0.737

Key Findings

1. Experience explains almost everything. years experience is the strongest numeric predictor ($r = 0.447$) and experience level the strongest categorical one (Cramér's $V = 0.337$). Mean salary climbs monotonically from \$62,582 at Entry to \$157,613 at Executive.

2. AI tool use does not pay a premium. Daily AI tool users average \$105,308 versus \$91,071 for non-users, but the association with salary band is weak ($V = 0.126$), and all three AI-related variables land in weak-to-negligible territory. The raw gap is largely confounded with seniority: senior people use AI tools more. **AI fluency neither pays you more nor prices you out. Experience does the work.**

3. Education matters far less than expected. education level scores $V = 0.092$, effectively noise. Within any seniority row of the experience $\times$ education pivot, salary varies far less across education columns than it does down the seniority axis.

4. The middle band is unlearnable with these features. Recall on the Medium class is 0.33–0.35 across all three models, with errors spreading evenly in both directions and AUC of just 0.575. Three different algorithm families failing identically is a **feature problem, not a model problem**, and it caps overall accuracy near 0.53.

5. Errors respect the ordinal structure. True High predicted as Low occurred only 2–7 times per model. Confusion concentrates between neighbouring bands, meaning the models get the direction right even when they miss the exact band.

Repository Structure

```
.  
├── data/  
│   └── ai_ds_job_salaries_2026.csv  
├── notebooks/  
│   └── M11_A8_Supervised_Learning_Classification.ipynb  
├── reports/  
│   └── A8_Supervised_Learning_Classification_Report.md  
├── requirements.txt  
└── README.md
```

Tech Stack

Python · pandas · NumPy · scikit-learn · SciPy · Matplotlib · Seaborn · Google Colab

Limitations

- **Sample size.** Analysis uses the first 1,000 rows; job title and industry have 12 and 10 levels respectively, leaving thin cells that inflate Cramér's V.
- **Accuracy ceiling.** ~53% is well short of production-grade. The dataset lacks the variables that most plausibly drive pay.
- **Fixed cut-points.** Band boundaries reflect 2026 percentiles and will drift as wages inflate.
- **Self-reported data.** Salary and satisfaction fields come from survey responses and carry the usual reporting bias.

Next Steps

- Add cost-of-living-adjusted location, company revenue/funding stage, and tenure at current employer, feature engineering is the highest-leverage improvement here, not further tuning
- Test an ordinal classifier or ordinal regression to exploit the natural Low < Medium < High ordering
- Deploy the pruned Decision Tree behind a FastAPI endpoint returning class probabilities, suppressing low-confidence predictions in favour of a group median
- Monitor population stability on incoming years experience and job_title, and recompute band cut-points quarterly

Author Ayleen Santander

Business Intelligence & Data Analytics | Willis College | Assignment 8: Supervised Learning Classification